Fixed Points, Linearization, and Its Limits

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Definition: Jacobian matrix

Let $\mathbf{F} = (f, g)$ be continuously differentiable near $\mathbf{z}^* = (x^*, y^*)$. The **Jacobian matrix of F at $\mathbf{z}^*$** is

$$D\mathbf{F}(\mathbf{z}^*) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}_{(x^*, y^*)}.$$

Notation: Displacement from an equilibrium

Let $\mathbf{z}^*$ be an equilibrium. For a nearby state $\mathbf{z}$, write

$$\mathbf{u} = \mathbf{z} - \mathbf{z}^*.$$

Observation: Taylor expansion at an equilibrium

Let $\mathbf{F}$ be twice continuously differentiable near an equilibrium $\mathbf{z}^*$. Taylor's theorem gives

$$\dot{\mathbf{u}} = J\mathbf{u} + \mathbf{R}(\mathbf{u}), \qquad J = D\mathbf{F}(\mathbf{z}^*),$$

where there are constants $C, \rho > 0$ such that

$$\|\mathbf{R}(\mathbf{u})\| \le C\|\mathbf{u}\|^2 \qquad \text{whenever } \|\mathbf{u}\| < \rho.$$

Equivalently,

$$\mathbf{R}(\mathbf{u}) = O\big(\|\mathbf{u}\|^2\big) \qquad (\mathbf{u} \to \mathbf{0}).$$

Definition: Linearized system

Let $\mathbf{z}^*$ be an equilibrium of $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$, and let $\mathbf{u} = \mathbf{z} - \mathbf{z}^*$. The **linearized system at $\mathbf{z}^*$** is

$$\dot{\mathbf{u}} = D\mathbf{F}(\mathbf{z}^*)\mathbf{u}.$$

Recall: Hyperbolic equilibrium

An equilibrium is **hyperbolic** if every eigenvalue λ of its Jacobian satisfies

$$\operatorname{Re}\lambda \neq 0.$$

Definition: Hyperbolic sink and source

A hyperbolic equilibrium is a **sink** if every eigenvalue of its Jacobian has negative real part. It is a **source** if every eigenvalue of its Jacobian has positive real part.

Recall: Homeomorphism

A **homeomorphism** is a continuous bijection with a continuous inverse.

Definition: Topological equivalence of phase portraits

Two local phase portraits are **topologically equivalent** if a homeomorphism maps trajectories of one portrait to trajectories of the other and preserves the direction of time.

Theorem: Hartman–Grobman theorem

Let $\mathbf{F}$ be continuously differentiable near a hyperbolic equilibrium $\mathbf{z}^*$. There are neighborhoods of $\mathbf{z}^*$ and $\mathbf{0}$ and a homeomorphism between them that maps $\mathbf{z}^*$ to $\mathbf{0}$ and maps local trajectory arcs of

$$\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$$

to local trajectory arcs of

$$\dot{\mathbf{u}} = D\mathbf{F}(\mathbf{z}^*)\mathbf{u}.$$

The map preserves the direction along each orbit while the arcs remain in the chosen neighborhoods. Consequently, a hyperbolic sink, source, or saddle has the same local orbit type as its linearization.

Idea: What linearization preserves

Hartman–Grobman preserves

$$\boxed{\text{local orbit structure}} \qquad \boxed{\text{time orientation}} \qquad \boxed{\text{sink/source/saddle type}}.$$

It does not assert equality of trajectories, preservation of distances, exact exponential rates, or any global basin statement.

Question: What does linearization say at three equilibria?

For

$$\dot{x} = -x + x^3, \qquad \dot{y} = -2y,$$

find and classify every equilibrium.

Solution. Equilibria satisfy

$$x(x^2 - 1) = 0, \qquad y = 0,$$

so they are

$$(0,0), \qquad (1,0), \qquad (-1,0).$$

The Jacobian is

$$J(x,y) = \begin{pmatrix} -1 + 3x^2 & 0 \\ 0 & -2 \end{pmatrix}.$$

At $(0,0)$, the eigenvalues are $-1, -2$, so it is an asymptotically stable node. At each of $(\pm 1, 0)$, the eigenvalues are $2, -2$, so it is a saddle. All three are hyperbolic; therefore the local classifications are valid for the nonlinear system.

Visualization: Exact product structure of the three-equilibrium system

The scalar x-flow is

$$\longleftarrow \ \circ_{-1} \ \longrightarrow \ \bullet_0 \ \longleftarrow \ \circ_1 \ \longrightarrow,$$

and

$$y(t) = y_0 e^{-2t}.$$

The lines

$$x = -1, \qquad x = 0, \qquad x = 1, \qquad y = 0$$

are invariant. In each vertical strip, the horizontal direction is the arrow from the scalar phase line and the vertical direction points toward $y = 0$.

Question: Can cubic terms change a linear center?

Let $a \in \mathbb{R}$ and consider

$$\dot{x} = -y + ax(x^2 + y^2), \qquad \dot{y} = x + ay(x^2 + y^2).$$

Classify the origin for every a.

Solution. The Jacobian at the origin is

$$J(0,0) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

with eigenvalues $\pm i$. The origin is nonhyperbolic, so linearization is inconclusive.
Let $r^2 = x^2 + y^2$. For $r > 0$,

$$\begin{aligned} r\dot{r} &= x\dot{x} + y\dot{y} \\ &= a(x^2 + y^2)^2 = ar^4, \end{aligned}$$

and

$$\begin{aligned} r^2\dot{\theta} &= x\dot{y} - y\dot{x} \\ &= x^2 + y^2 = r^2. \end{aligned}$$

Question: Can cubic terms change a linear center? (continued)

Thus

$$\dot{r} = ar^3, \qquad \dot{\theta} = 1.$$

Therefore

$$\begin{array}{r|c} a < 0 & \text{asymptotically stable spiral} \\ a = 0 & \text{center} \\ a > 0 & \text{unstable spiral} \end{array}$$

even though the linearization is the same in all three cases.

Question: What is the exact radial behavior?

Solve

$$\dot{r} = ar^3, \qquad r(0) = r_0 > 0.$$

Solution. Separation gives

$$\int_{r_0}^{r(t)} r^{-3}\,\mathrm{d}r = \int_0^t a\,\mathrm{d}s,$$

so

$$-\frac{1}{2r(t)^2} + \frac{1}{2r_0^2} = at.$$

Hence

$$r(t) = \frac{r_0}{\sqrt{1 - 2ar_0^2 t}}.$$

If $a < 0$, then $r(t) \to 0$ algebraically as $t \to \infty$. If $a > 0$, the solution becomes unbounded at

$$T_{\max} = \frac{1}{2ar_0^2}.$$

Thus the phrase "unstable spiral" is local; for $a > 0$ the nonzero solution does not exist for all $t \geq 0$.

Visualization: When is linearization decisive?

spectrum	status	next step
$\operatorname{Re}\lambda_j \neq 0$ for all j	hyperbolic	use Hartman–Grobman
$\lambda = \pm i\omega,\ \omega > 0$	nonhyperbolic	linearization is inconclusive; analyze the nonlinear system
0 is an eigenvalue	nonhyperbolic	analyze the nonlinear dynamics in the zero-eigenvalue direction

Observation: Repeated real eigenvalues can be hyperbolic

A planar equilibrium whose Jacobian has the repeated real eigenvalue $\lambda \neq 0$ is hyperbolic. Whether the eigenspace is one- or two-dimensional changes the linear geometry, but the sign of λ still determines whether the equilibrium is a sink or a source.

Definition: Locally invariant graph

Consider
$$\dot{u} = F(u,v), \qquad \dot{v} = G(u,v),$$
and let $I \subseteq \mathbb{R}$ be an interval on which $h : I \to \mathbb{R}$ is differentiable. The graph
$$\Gamma = \{(u, h(u)) : u \in I\}$$
is **locally invariant near a point** $p \in \Gamma$ if there is an open neighborhood N of p such that every solution starting in $\Gamma \cap N$ remains in Γ for as long as it remains in N.

Proposition: Invariant graph equation

If the differentiable graph $v = h(u)$ is locally invariant near every point of the graph for
$$\dot{u} = F(u,v), \qquad \dot{v} = G(u,v),$$
then
$$h'(u)F(u,h(u)) = G(u,h(u)) \qquad (u \in I).$$

Proof. Along a trajectory on the graph,
$$\dot{v} = h'(u)\dot{u},$$
and substitution of the differential equations gives the result.

Question: How does a nonlinear term bend an invariant direction?

Consider
$$\dot{x} = x - xy, \qquad \dot{y} = -2y + x^2.$$
Classify the origin. Assuming its local unstable invariant curve is C^3 and tangent to the x-axis, find its quadratic term.

Solution.
$$J(x,y) = \begin{pmatrix} 1-y & -x \\ 2x & -2 \end{pmatrix}, \qquad J(0,0) = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}.$$
The eigenvalues are 1 and -2, so the origin is a hyperbolic saddle. Since $x = 0$ implies $\dot{x} = 0$, the y-axis is the stable invariant curve.
Write the unstable curve as
$$y = h(x) = cx^2 + O(x^3) \qquad (x \to 0).$$
The invariance equation is
$$h'(x)x(1 - h(x)) = -2h(x) + x^2.$$
Since
$$h'(x) = 2cx + O(x^2) \qquad (x \to 0),$$
$$2cx^2 + O(x^3) = (1-2c)x^2 + O(x^3) \qquad (x \to 0).$$
Thus
$$2c = 1 - 2c, \qquad c = \frac{1}{4},$$

Question: How does a nonlinear term bend an invariant direction? (continued)

and

$$y = \frac{1}{4}x^2 + O(x^3) \qquad (x \to 0).$$